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-----SIMBOLOS E FORMULAS-----

$\approx$

$$H_0 \approx 2.2 \times 10^{-18} \, s^{-1}$$

$$\Sigma \star(R)=(M/L)I(R)$$

$$\Sigma \star(R)$$

$$I(R)$$

$$(\sqrt{G\sqrt{\pi\rho}})^*\,M^{1/4}$$

$$\frac{\rho}{\rho_t}$$

$$V=c\, \Big(\frac{1}{c}\sqrt{\frac{MG}{R}}+1\Big)\,e^{H_0T_0*(1-\frac{\rho}{\rho_t})}\,\,\,\,\,\boldsymbol{-1)}$$

- $\pi \rightarrow 03C0 + Alt+X$
- $\Pi \rightarrow 03A0 + Alt+X$
- Equação $\rightarrow \backslash pi$
-
-
- $\rho \rightarrow 03C1 + Alt+X$

$$\cdot \quad \mathbf{P} \rightarrow \mathbf{03A1} + \mathbf{Alt+X}$$

$$\cdot \quad \text{Equação} \rightarrow \backslash \text{rho}$$

$$\rho$$

$$\rho$$

$$\mathbf{V=c} \left(\left(\frac{1}{c} \sqrt{\frac{MG}{R}} + \mathbf{1} \right) e^{H_0 T_0 * (1-\frac{K}{K_T})} \right. - \mathbf{1}) \quad \mathbf{[F25]}$$

$$\backslash \text{rho}$$

$$_rho \backslash rho$$

$$\propto$$

$$\mathbf{V_t=c} \left(\left(\frac{1}{c} \sqrt{\frac{MG}{R}} + \mathbf{1} \right) e^0 \right. - \mathbf{1}) \quad \mathbf{[F25]}$$

$$\mathbf{V_t=}\left(\frac{1}{c}\sqrt{\frac{MG}{\sqrt{M/K_2}}}\right)$$

$$\mathbf{V_t=c} \left(\left(\frac{1}{c} \sqrt{\frac{MG}{R}} + \mathbf{1} \right) e^0 \right. - \mathbf{1})$$

$$\text{Apparent Distance Formula}$$

The measured distance is inversely proportional to the length of the measurement pattern. We can synthesize this idea mathematically with the following formula (See the **Appendix A**):

$$D(\Delta t) = D_0 * F_j(\Delta t) \quad (E4)$$

(Distance Formula with the Jocaxian Factor).

Where:

"t₀" is the time at which the photon was emitted by the galaxy.

"t" is the time at which Earth received this photon.

"D₀" is the distance we would measure, from Earth to galaxy at time "t₀"

"D(Δt)" is the distance we measured from Earth to a galaxy after "Δt" time.

"Δt" = t - t₀ Time period

As F_j(Δt) grows exponentially with time (E3) the Earth's distance from the galaxy will also appear to increase exponentially with time.

≡

Hubble's Law

With the formula of the apparent distance (E4) we can calculate the apparent distance speed:

$$V = d[D(\Delta t)] / dt = (E5-A)$$

$$V = d[D_0 * \exp(\ln(2) * \Delta t / T_j)] / dt \quad (E5-B)$$

$$V = (\ln(2) / T_j) * D(\Delta t) \quad (E6)$$

(Distant galaxies' apparent distance speed formula.)

But **Hubble's law** is exactly like this:

$$V = H_0 * D \quad (E7)$$

(Hubble's Law)

Where (H_0 = Hubble's Constant and D is the distance from the galaxy)

As $E6 = E7$, now we can determine T_j :

$$T_j = \ln(2) / H_0 \quad (E8)$$

Substituting (E8) into (E3) we will have:

$$F_j(\Delta t) = \exp(H_0 * \Delta t) \quad (E9)$$

(Jocaxian Factor in terms of the Hubble constant)

What provides us:

$$D(\Delta t) = D_0 * \exp(H_0 * \Delta t) \quad (E10)$$

(Apparent distance formula in terms of the Hubble constant)

Where

$$J_0 = \frac{H_0 R_t^2}{M_t} \quad (F7)$$

(Jocax Constant $\approx 1,37E-18 \text{ m}^2 \cdot \text{s}^{-1} \cdot \text{kg}^{-1}$).

Where

H_0 = Hubble Constant ($\approx 2,2E-18 \text{ s}^{-1}$)

R_t = Distance from Earth to center of Milk Way ($\approx 2,5E20 \text{ m}$)

M_t = Mass of Milk Way ($\approx 1E41 \text{ Kg}$)

5.1 Core Hypothesis

The gravitational field (g) contracts space **continuously**, with the rate depending on g 's intensity. From [\[01\]](#), the contraction formula is:

$$L(t) = \frac{L_0}{2^{\Delta t/T_j}} \quad (F1)$$

(Contraction formula from the point of view of a non-shrinking observer)

Where:

$L(t)$ = Measure of L_0 (decreasing) by a "Sidereal Observer".

t_0 =Initial time (arbitrary)

L_0 =Length measured in $t=t_0$

T_j = Jocaxian Time (Time to shrink to half)

$\Delta t = t - t_0$

Rewritten using the *Jocaxian Factor* (F_j):

$$F_j(\Delta t) = 2^{\Delta t/T_j} \quad (F2)$$

Jocaxian's Factor

We can also rewrite the same Jocaxian Factor (F_j) in a more friendly way:

$$F_j(\Delta t) = e^{\left(\ln(2) * \frac{\Delta t}{T_j}\right)} \quad (F3)$$

We can rewrite (F1):

$$L = L_0 / F_j(\Delta t) \quad (F4)$$

5.2 Gravitational Dependence

We know that the rate of contraction depends on the gravitational field (g). If the gravitational field increases, the contraction will be faster, that is, the time for space to fall by half decreases.

Therefore, we will include a new hypothesis:

The contraction time is inversely proportional to the intensity of the gravitational field.

So, we have:

$$T_j(g) = T_j \cdot \frac{g_t}{g} \quad (\text{F5})$$

Where

T_j = Jocaxian Time (Time to Length contract to half)

$g_t = GM_t/R_t^2$ (gravitational field at Earth (Milk Way)).

$g = GM/R^2$ (gravitational field at distance R from galactic center).

This leads to the **generalized Jocaxian Factor**:

$$F_j(\Delta t, M, R) = e^{\frac{J_0 M \Delta t}{R^2}} \quad (\text{F6})$$

Where

$$J_0 = \frac{H_0 R_t^2}{M_t} \quad (\text{F7})$$

(Jocax Constant $\approx 1,37E-18 \text{ m}^2 \cdot \text{s}^{-1} \cdot \text{kg}^{-1}$).

Where

H_0 = Hubble Constant ($\approx 2,2E-18 \text{ s}^{-1}$)

R_t = Distance from Earth to center of Milk Way ($\approx 2,5E20 \text{ m}$)

M_t = Mass of Milk Way ($\approx 1E41 \text{ Kg}$)

6. Galactic Rotation Curves

In the Milky Way, the photon after T_0 will have a shorter wavelength:

$$\lambda^T_0 = \lambda_0 / F^T_J(T_0) = \lambda_0 / \exp(H_0 T_0)$$

Where:

λ_0 = Wave length at beginning

λ^T_0 = wave length of photon in Earth after T_0

F^T_J = Jocaxian Factor on Earth

T_0 = Time from beginning of Galaxy until Now (13G Years)

In Messier_33 the contraction rate is different, and we must use the general F_j :

$$\lambda^{M_0} = \lambda_0 / F_j(T_0, M, R)$$

Where:

λ^{M_0} = wave length of photon in Messier_33 Earth **after** T_0

$F_j(T_0, M, R)$ = generalized Jocaxian Factor

M = Messier_33 Mass ($1.4E39$ Kg)

R = Distance of star to center of Messier_33

However, the Messier_33 star that emits the photon, will present a redshift that depends on its *rotation speed* .:

$$V = cz$$

Where c = Speed of light and z = redshift

However

$$V = \text{SQRT}(MG/R)$$

$$z = V/c = (\lambda^{M_1} / \lambda^{M_0} - 1)$$

Where

λ^{M_1} = wave length of photon in Messier33 due rotation velocity

$$\lambda^{M_1} = \lambda^{M_0} (\text{SQRT}(MG/R)/c + 1)$$

The final redshift relative to the Milky Way will be:

$$Z_f = (\lambda^{M_1} - \lambda^T_0) / \lambda^T_0$$

So,

$$Z_f = (\lambda^{M_0} (\text{SQRT}(MG/R)/c + 1)) / \lambda^T_0 - 1$$

Replacing we will have:

$$Z_f = (\exp(H_0 T_0) / F_j(T_0, M, R)) (\text{SQRT}(MG/R)/c + 1) - 1$$

But $v = cz$, then **the speed measured on Earth** (without considering the effect of “dark energy” = Our contraction during the photon’s journey), will be:

$$V = c \{ (\exp(H_0 T_0) / F_j(T_0, M, R)) (\text{SQRT}(MG/R)/c + 1) - 1 \}$$

Substituting $F_j(T_0, M, R)$ we will finally have the galaxy's rotation curve:

$$V_t = c \left(\left(\frac{1}{c} \sqrt{\frac{MG}{R}} + 1 \right) e^{H_0 T_0 - \frac{J_0 M T_0}{R^2}} - 1 \right)$$

Galaxy Rotation Curve (Apparent Velocity)

Where:

C = Speed of Light ($3E8$ m/s)

M = Mass of the Emitting Galaxy ($1,4E40$ Kg)

G = Gravitational Constant ($6.7E-11$)

R = Distance from the Star to the center of the galaxy

H_0 = Hubble Constant ($2.2E-18$)

T_0 = Contraction Time (13 billion years = $4E16$ s)

J_0 = Jocas Constant ($1,37E-18$)

6.2-For the galaxy Messier 33

Substituting the constant values we will have the Speed of the Messier Galaxy Star Observed from Earth:

$$V_t = 3E5 \left(\left(\sqrt{\frac{2,4E6}{R}} + 1 \right) e^{(0,088 - \frac{7,7E38}{R^2})} - 1 \right)$$

$$Measurement = \overline{(numerical\ value)} \times (unit\ of\ measurement)$$

\overbar{

Equilíbrio entre a Dilatação Temporal Gravitacional (TR) e a Escala Temporal da DUT

Nesta seção, estimamos a escala de tempo necessária para que a variação temporal prevista pela Teoria do Universo Decrescente (DUT) compense a dilatação temporal gravitacional prevista pela Teoria da Relatividade Geral (TR) na superfície da Terra.

De acordo com a TR, o tempo próprio medido a uma distância radial r de uma massa M , em relação a um observador distante (comóvel), é dado por:

$$\frac{T_{\text{Terra}}}{T_{\text{comóvel}}} = \sqrt{1 - \frac{2GM}{rc^2}}$$

Esse fator é menor que a unidade, indicando que relógios em um campo gravitacional correm mais lentamente em relação a um observador distante.

Em contraste, no contexto da DUT, a variação temporal em relação ao referencial comóvel é dada por:

$$\frac{T_{\text{Terra}}}{T_{\text{comóvel}}} = e^{H_0 \Delta t}$$

onde H_0 é a constante de Hubble e Δt é o intervalo de tempo cosmológico.

Assumindo que ambos os efeitos atuam simultaneamente e de forma independente, o fator temporal combinado é dado pelo produto:

$$e^{H_0 \Delta t} \sqrt{1 - \frac{2GM}{rc^2}}$$

A condição para compensação exata entre os dois efeitos é, portanto:

$$e^{H_0 \Delta t} \sqrt{1 - \frac{2GM}{rc^2}} = 1$$

Resolvendo para Δt , obtemos:

$$\Delta t = -\frac{1}{2H_0} \ln\left(1 - \frac{2GM}{rc^2}\right)$$

Para campos gravitacionais fracos ($\frac{2GM}{rc^2} \ll 1$), utilizamos a aproximação:

$$\ln(1 - x) \approx -x$$

obtendo:

$$\Delta t \approx \frac{GM}{rc^2 H_0}$$

Substituindo os parâmetros físicos da Terra:

$$\begin{aligned} G &= 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}, \\ M &= 5.972 \times 10^{24} \text{ kg}, \\ r &= 6.371 \times 10^6 \text{ m}, \\ c &= 2.998 \times 10^8 \text{ m/s} \end{aligned}$$

e adotando:

$$H_0 \approx 2.2 \times 10^{-18} \text{ s}^{-1}$$

obtemos:

$$\frac{GM}{rc^2} \approx 6.96 \times 10^{-10}$$

e, portanto:

$$\Delta t \approx 3.16 \times 10^8 \text{ s} \approx 10 \text{ anos}$$

Interpretação

Esse resultado indica que, no contexto da DUT, o efeito cumulativo da variação temporal torna-se comparável em magnitude à dilatação temporal gravitacional na superfície da Terra após aproximadamente 10 anos.

Em outras palavras, enquanto a dilatação temporal na TR é um efeito local fixo determinado pelo potencial gravitacional, a DUT prevê uma evolução temporal contínua. Após um intervalo cosmológico suficientemente longo, a contribuição da DUT pode compensar efetivamente o efeito relativístico quando ambos são comparados em relação a um referencial comóvel.

=====

Você pode escrever assim:

$$e^{H_0 \Delta t} \sqrt{1 - \frac{2GM}{rc^2}} = 1$$

de onde segue:

$$\Delta t = -\frac{1}{2H_0} \ln \left(1 - \frac{2GM}{rc^2} \right) \approx \frac{GM}{rc^2 H_0}$$

Para a Terra:

$$\Delta t \approx 3.16 \times 10^8 \text{ s} \approx 10 \text{ anos}$$

Balance Between Gravitational Time Dilation (GR) and DUT Temporal Scaling

In this section, we estimate the timescale required for the temporal scaling predicted by the Decreasing Universe Theory (DUT) to compensate for the gravitational time dilation predicted by General Relativity (GR) at the Earth's surface.

According to GR, the proper time measured at a radial distance r from a mass M , relative to a distant (comoving) observer, is given by:

$$\frac{T_{\text{Earth}}}{T_{\text{comoving}}} = \sqrt{1 - \frac{2GM}{rc^2}}$$

This factor is smaller than unity, indicating that clocks in a gravitational field run more slowly relative to a distant observer.

In contrast, within the DUT framework, the temporal scaling relative to the comoving frame is given by:

$$\frac{T_{\text{Earth}}}{T_{\text{comoving}}} = e^{H_0 \Delta t}$$

where H_0 is the Hubble constant and Δt is the cosmological time interval.

Assuming that both effects act simultaneously and independently, the combined temporal factor is given by the product:

$$e^{H_0 \Delta t} \sqrt{1 - \frac{2GM}{rc^2}}$$

The condition for exact compensation between the two effects is therefore:

$$e^{H_0 \Delta t} \sqrt{1 - \frac{2GM}{rc^2}} = 1$$

Solving for Δt , we obtain:

$$\Delta t = -\frac{1}{2H_0} \ln \left(1 - \frac{2GM}{rc^2} \right)$$

For weak gravitational fields ($\frac{2GM}{rc^2} \ll 1$), we use the approximation:

$$\ln(1 - x) \approx -x$$

leading to:

$$\Delta t \approx \frac{GM}{rc^2 H_0}$$

Substituting the physical parameters of the Earth:

$$G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}, \\ M = 5.972 \times 10^{24} \text{ kg}, r = 6.371 \times 10^6 \text{ m}, c = 2.998 \times 10^8 \text{ m/s}$$

and adopting:

$$H_0 \approx 2.2 \times 10^{-18} \text{ s}^{-1}$$

we obtain:

$$\frac{GM}{rc^2} \approx 6.96 \times 10^{-10}$$

and therefore:

$$\Delta t \approx 3.16 \times 10^8 \text{ s} \approx 10 \text{ years}$$

Interpretation

This result indicates that, within the DUT framework, the cumulative temporal scaling becomes comparable in magnitude to the gravitational time dilation at the Earth's surface over a timescale of approximately 10 years.

In other words, while gravitational time dilation in GR is a fixed local effect determined by the gravitational potential, the DUT predicts a continuous temporal evolution. After a sufficiently long cosmological interval, the DUT contribution can effectively compensate for the relativistic effect when both are compared with respect to a comoving reference frame.



$$\overline{\text{espaço}} = \overline{Z}$$

$$\overline{L}_0$$

$$\overline{L}_t$$

$$\frac{S_t}{S_0} = \frac{s_t}{s_0}$$

$$\overline{T}_0 \text{ dfdfdfdf}$$

$$\overline{T}_t$$

$$\overline{Z}$$

$$\exp(H_0 * \Delta t)$$

$$\exp(H_0 * \Delta t) \text{ (Fator de contração)}$$

$$\exp(H_0 * \Delta t) \text{ (Fator de contração)}$$

aaaaa

$$S_t/S_0 = (\overline{L}_0 / \overline{C}_0) * (\overline{C}_t / \overline{L}_t)$$

$$\text{Mas [] } L_t = L_0 * \overline{Z}$$

Segue que :

$$S_t/S_0 = \overline{C}_t / \overline{C}_0 * (1/\overline{Z})$$

O símbolo de “aproximadamente” é:

[
\approx
]

✔ Como fazer no Word

Método 1 (equação — recomendado)

1. Pressione:

Alt + =

2. Digite:

\approx

3. Pressione espaço

Método 2 (atalho rápido)

Digite:

2248

e depois pressione:

Alt + X

✔ Outros símbolos parecidos (úteis)

Símbolo Comando Significado

≈	\approx	aproximadamente
≈	\simeq	aproximadamente igual (mais formal)
~	\sim	da ordem de / proporcional
≅	\cong	aproximadamente congruente

Dica para seu artigo

- Use ≈ (**\approx**) para valores numéricos
 - Use ~ (**\sim**) para ordem de grandeza
-

Se quiser, posso montar uma lista só com símbolos matemáticos úteis pro seu paper
👍

Use o underscore _

Digite:

T_0

Pressione **espaço**

👉 Resultado:

$$\frac{T_0}{A_0/B_0}$$

$$\overline{\Delta t = 0 \rightarrow Z = 1 \rightarrow m_0 = m_t}$$

$$\Delta t = 0 \rightarrow \overline{Z} = 1 \rightarrow m_0 = m_t$$

Exemplo com fração:

$$\frac{T_{\text{obs}}}{T_{\text{emit}}}$$

👉

$$\frac{T_{obs}}{T_{emit}}$$

Exemplo com expoente + subscripto:

$$x_i^2$$

👉

$$x_i^2$$

$$\lambda = \lambda_0 \cdot \exp(H_0 \cdot \Delta t) \text{ [F03]}$$

Wavelength Measured within a Gravitational Field

Where:

λ = Wavelength measured when the photon *reaches* Earth.
 λ_0 = Wavelength when the photon *leaves* the galaxy, at T_0 .

We'll also use the redshift (Z) setting:

$$Z = (\lambda - \lambda_0) / \lambda_0 \text{ [F04]}$$

Redshift Definition

Utilizando [FR03] e [FT18] acima, derivamos o “tempo de dilatação cosmológica”
[05][06][07]:

$$\frac{\overline{T}_0}{\overline{T}_t} = Z+1 \quad [\text{FR04}]$$

COSMOLOGICAL TIME DILATATION

Digite a equação aqui.

$$\frac{\overline{T}_0}{\overline{T}_t} = (Z+1) \overline{T}_t \quad [\text{FR04}]$$

COSMOLOGICAL TIME DILATATION

Ou seja a medida de tempo no referencial comoving

$$\overline{T}_0/\overline{T}_t$$

 Método via menu (alternativo)

1. Inserir → Equação
2. Índices → escolha subscrito
3. Preencha os campos

 Dica importante (para seu paper)

Sempre use:

- T_0, T_t, S_0, S_t

- $\backslash\lambda\text{bda_0, c_t}$

👉 Isso deixa consistente e profissional

Se quiser, posso te passar um **padrão completo de notação para seu artigo (tipo guia de estilo científico)** 👍

Aproximadamente = $\approx \approx$

$\overline{(X+Y)}$ espaço = $\overline{Z + Y}$

$\overline{(X)}$ espaço = $\overline{X}$ $\overline{x}$

$\overline{(Z)}$ espaço = $\overline{Z}$

$\overline{(c)}$ espaço = $\overline{c_0}$ $\overline{c_t}$

$x/y = \frac{x}{Y}$ $\times \frac{A}{B}$

A/B espaço = $\frac{A}{B}$

N_c

X_0

$m_t = \overline{Z} * m_0$

$\overline{(t)}$ $\overline{T}$

$\overline{(\backslash\lambda\text{bda}}$ espaço) espaço = $\overline{\lambda}$

C_0 $\overline{c_0}$

S_0

$\lambda_0 = \overline{\lambda_0} * m_0$

$c_0 = \overline{c_0} * m_0 / S_0$

$$\overline{c}_0=N_c*\overline{\lambda}_0$$

$$\overline{c}_t\,=\,N_c*\,\overline{\lambda}_t$$

$$\overline{\lambda}_0=\overline{\lambda}_t$$

$$L_o = \overline{L}_0 * m_0$$

$$S_0$$

$$S0$$

$$\backslash \mathrm{bar}\{c\}$$

$$(a+b)/(c+d)\,\mathrm{espaco}\,=\,\frac{a+b}{c+d}$$

$$T_2=T_1\,e^{\{H_0\,\Delta\,t\}}$$

$$1+z=e^{\{H_0\,\Delta\,t\}}$$

$$\frac{T_2}{T_1}=1+z$$

$$c=\frac{\lambda}{T}$$

$$\lambda=\lambda_0\,e^{\{-H_0\,\Delta\,t\}}$$

$$T=T_0\,e^{\{-H_0\,\Delta\,t\}}$$

$$c=\frac{\lambda_0}{T_0}$$

$$T_{\mathrm{RG}}=\frac{T_0}{\sqrt{1-\frac{2GM}{rc^2}}}$$

$$\frac{T'}{T_0} = \frac{1+z}{\sqrt{1 - \frac{2GM}{rc^2}}}$$

$$\lambda = \frac{\lambda_0}{e^{H_0 \Delta t}}$$

$$\Delta t$$

$$\Omega$$

Sempre use:

- λ (não escreva “lambda”)
 - Δt (não “dt” simples)
 - $\frac{\quad}{\quad}$ (frações bonitas)
 - $e^{\dots}$ (exponencial correto)
-

$$t = \frac{1}{H_0} \ln(1+z)$$

$$L(t) = L_0 / 2^{\Delta t / T_j} \quad (E1)$$

(Formula of space contraction from the point of view of an "SO")

Where:

$L(t)$ = Measure of L_0 in the "LS" by a "Sidereal Observer".

t_0 = Initial time (arbitrary)

L_0 = Length measured in $t=t_0$

T_j = Jocaxian Time

$\Delta t = t - t_0$

$$F_j(\Delta t) = 2^{\Delta t / T_j} \quad (E1-B)$$

(Jocaxian Factor).

We can rewrite (E1):

$$L = L_0 / F_j(\Delta t) \quad (E2)$$

We can also rewrite the same Jocaxian Factor (F_j) in a more friendly way:

$$F_j(\Delta t) = \exp(\ln(2) * \Delta t / T_j) \quad (E3)$$

$$D(\Delta t) = D_0 * F_j(\Delta t) \quad (E4)$$

Where:

" t_0 " is the time at which the photon was emitted by the galaxy.

" t " is the time at which Earth received this photon.

" D_0 " is the distance we would measure, from Earth to galaxy at time " t_0 "

" $D(\Delta t)$ " is the distance we measured from Earth to a galaxy after " Δt " time.

" Δt " = $t - t_0$ Time period

As $F_j(\Delta t)$ grows exponentially with time (E3)

$$T_j = \ln(2) / H_0 \quad (E8)$$

Substituting (E8) into (E3) we will have:

$$F_j(\Delta t) = \exp(H_0 * \Delta t) \quad (E9)$$

(Jocaxian Factor in terms of the Hubble constant)

What provides us:

$$D(\Delta t) = D_0 * \exp(H_0 * \Delta t) \quad (E10)$$

(Apparent distance formula in terms of the Hubble constant)

If we want to calculate the real distance from Earth to the galaxy, using the measurements that our scales had at the time the photon was emitted (at $t = t_0$) then:

$$\Delta t = D_0/c \quad (E11)$$

(Time for a photon emitted from the galaxy to reach us, where c = speed of light)

From (E11) and (E10) we will have:

$$D = D_0 \exp(D_0 * H_0/c) \quad (E12)$$

(Apparent distance of a galaxy depending on the actual distance)

$$\text{Dark Matter} = M - M_0 = M_0 * (\exp(3 * D_0 * H_0/c) - 1) \quad (E27)$$

$$L = L_0 / F_j(\Delta t)$$

Onde :

L_0 = Tamanho medido pelo observador comóvel (que não sofre contração)

$F_j(\Delta t)$ = $\exp(H_0 * \Delta t)$ Fator Jocabiano de contração do espaço com o tempo.

H_0 = Contante de Hubble

$$L_0 = \lambda_{0c}$$

$$L = \lambda_c \quad (\text{após a contração})$$

$$\lambda_c = \lambda_{0c} / \exp(H_0 * \Delta t)$$

$$N_2 = N_1 \exp(H_0 \Delta t)$$

is:

$$L(t) = \frac{L_0}{2^{\Delta t/T_j}} \quad (F1)$$

(Contraction formula from the point of view of a non-shrinking observer)

Where:

$L(t)$ = Measure of L_0 (decreasing) by a "Sidereal Observer".

t_0 =Initial time (arbitrary)

L_0 =Length measured in $t=t_0$

T_j = Jocaxian Time (Time to shrink to half)

$\Delta t = t - t_0$

$$Z+1 = \exp(H_0 * T) \quad [F05]$$

Rewritten using the *Jocaxian Factor* (F_j):

$$F_j(\Delta t) = 2^{\Delta t / T_j} \quad (F2)$$

Jocaxian's Factor

We can also rewrite the same Jocaxian Factor (F_j) in a more friendly way:

$$F_j(\Delta t) = e^{\left(\ln(2) * \frac{\Delta t}{T_j} \right)} \quad (F3)$$

We can rewrite (F1):

$$L = L_0 / F_j(\Delta t) \quad (F4)$$

So, we have:

$$T_j(g) = T_j \cdot \frac{gt}{g} \quad (F5)$$

Where

T_j =Jocaxian Time (Time to Length contract to half)

$g_t = GM_t/R_t^2$ (gravitational field at Earth (Milk Way)).

$g = GM/R^2$ (gravitational field at distance R from galactic center).

This leads to the **generalized Jocaxian Factor**:

$$F_j(\Delta t, M, R) = e^{\frac{J_0 M \Delta t}{R^2}} \quad (\text{F6})$$

Where

$$J_0 = \frac{H_0 R_t^2}{M_t} \quad (\text{F7})$$

(**Jocax Constant** $\approx 1,37E-18 \text{ m}^2 \cdot \text{s}^{-1} \cdot \text{kg}^{-1}$).

Where

H_0 =Hubble Constant ($\approx 2,2E-18 \text{ s}^{-1}$)

R_t = Distance from Earth to center of Milk Way ($\approx 2,5E20 \text{ m}$)

M_t = Mass of Milk Way ($\approx 1E41 \text{ Kg}$)

Substituting $F_j(T_0, M, R)$ we will finally have the galaxy's rotation curve:

$$V_t = c \left(\frac{1}{c} \sqrt{\frac{MG}{R}} + 1 \right) e^{H_0 T_0 - \frac{J_0 M T_0}{R^2}} - 1) \quad -1)$$

Galaxy Rotation Curve (Apparent Velocity)

Where:

C = Speed of Light ($3E8 \text{ m/s}$)

M = Mass of the Emitting Galaxy ($1,4E40 \text{ Kg}$)

G = Gravitational Constant ($6.7E-11$)

R = Distance from the Star to the center of the galaxy

H_0 = Hubble Constant ($2.2E-18$)

T_0 = Contraction Time (13 billion years = $4E16 \text{ s}$)

J_0 = Jocax Constant ($1,37E-18$)

Galaxy Star Observed from Earth:

$$V_t = 3E5 \left(\sqrt{\frac{2,4E6}{R}} + 1 \right) e^{(0,088 - \frac{7,7E38}{R^2})} - 1$$

Podemos estimar a taxa da variação do tempo aqui na Terra em relação ao observador comóvel:

Taxa de variação Temporal

Chamemos de $\Delta t'$ a taxa de variação do tempo depois de um tempo cosmológico Δt medido no referencial comoving. Então teremos:

$$\Delta t' = \frac{T_2 - T_1}{T_1} = \frac{T_1 \exp(H_0 \Delta t) - T_1}{T_1} = \exp(H_0 * \Delta t) - 1$$

$$\Delta t' = \exp(H_0 * \Delta t) - 1 \quad [F7]$$

Taxa da contração temporal do tempo no ambiente gravitacional

Como exemplos podemos estimar a variação temporal para diferentes tempos cosmológicos e distâncias correspondentes:

Alguns Valores Numéricos

Se Tomarmos :

$c/H_0 = 14,3$ Bilhões de anos

Podemos montar a seguinte tabela :

Redshift (Z)	Universo Decrescente Distância (Comoving) (em bilhões de anos-luz)	Variação Temporal $\Delta t'$ (em %)	Diferença
0.0001			
0.001			
0.01			
0.1			
0.5			
1	9.9		
2	15.7		
3	19.8		
4	23.0		
10	34.2		

Na Teoria da Relatividade Geral, a dilatação temporal em um campo gravitacional pode ser escrita como:

$$T_{RG} = \frac{T_0}{\sqrt{1 - \frac{2GM}{rc^2}}}$$

onde T_0 é o tempo medido por um observador distante e T_{RG} o tempo medido no ambiente gravitacional.

Pela DUT, mostramos anteriormente que a variação temporal associada à contração das escalas físicas é dada por:

$$T_{DUT} = T_0 \exp(H_0 \Delta t)$$

Se considerarmos que ambos os efeitos atuam simultaneamente, a variação temporal total pode ser escrita como:

$$T' = T_0 \cdot \frac{1}{\sqrt{1 - \frac{2GM}{rc^2}}} \cdot \exp(H_0 \Delta t)$$

ou:

$$\frac{T'}{T_0} = \frac{\exp(H_0 \Delta t)}{\sqrt{1 - \frac{2GM}{rc^2}}}$$

Mostrando que a dilatação do tempo passa a depender tanto do campo gravitacional local quanto do tempo cosmológico acumulado.

Na DUT, em relação ao referencial comóvel não contraído, os processos físicos em um ambiente gravitacional transcorrem progressivamente mais rapidamente ao longo do tempo cosmológico, uma vez que a unidade temporal local diminui continuamente.

$$c = \frac{\lambda}{T}$$

$$\frac{\lambda_2}{T_2} = \frac{\lambda_1}{T_1}$$

Em nosso caso particular : $L_0 = \lambda_{0c}$ (não contraído) e
 $L = \lambda_c$ (após a contração)

$$c_1 = \frac{\lambda_{0c}}{T_1} = \frac{\lambda_{0c}}{T_1}$$

$$c_2 = \frac{\lambda_c}{T_2} = \lambda_{0c} \exp(-H_0 \Delta t)$$

$$c = \frac{\lambda_0(1+Z)}{T_0(1+Z)} = \frac{\lambda_0}{T_0}$$

Segue um trecho **pronto para inserir no artigo**, com dedução clara, valores numéricos e linguagem adequada:

Numerical Comparison Between Gravitational Time Dilation (GR) and DUT

In this section, we compare the magnitude of gravitational time dilation predicted by General Relativity (GR) at the surface of the Earth with the temporal variation predicted by the Decreasing Universe Theory (DUT), and determine the DUT scale (in terms of time, distance, and redshift) required to reproduce the same effect.

Gravitational Time Dilation at the Earth's Surface

According to General Relativity, the relation between the proper time $d\tau$ measured at a radial distance r from a mass M , and the coordinate time dt measured by an observer far from the gravitational field, is given by:

$$d\tau = dt \sqrt{1 - \frac{2GM}{rc^2}}$$

For weak gravitational fields, such as that of the Earth, we can use the approximation:

$$d\tau \approx dt \left(1 - \frac{GM}{rc^2}\right)$$

Thus, the fractional time difference is:

$$\frac{dt - d\tau}{dt} \approx \frac{GM}{rc^2}$$

Using the known physical constants:

$$G = 6.674 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$$

$$M = 5.972 \times 10^{24} \text{ kg}$$

$$r = 6.371 \times 10^6 \text{ m}$$

$$c = 2.998 \times 10^8 \text{ m/s}$$

we obtain:

$$\frac{GM}{rc^2} \approx 6.96 \times 10^{-10}$$

Therefore, for a time interval of $dt = 1$ second measured by a distant observer, the proper time at the Earth's surface is:

$$d\tau \approx 1 - 6.96 \times 10^{-10} \text{ s}$$

This corresponds to a time difference of:

$$\Delta t_{\text{GR}} \approx 6.96 \times 10^{-10} \text{ s per second}$$

Over longer timescales:

- per day:

$$\approx 6.0 \times 10^{-5} \text{ s } (\sim 60 \mu\text{s})$$

- per year:

$$\approx 2.2 \times 10^{-2} \text{ s}$$

Equivalent Effect in the DUT Framework

In the DUT framework, the temporal scaling is given by:

$$\frac{T}{T_0} = e^{H_0 \Delta t} = 1 + z$$

To reproduce the same fractional variation observed in GR, we impose:

$$e^{H_0 \Delta t} - 1 = 6.96 \times 10^{-10}$$

For small values, we use:

$$\ln(1 + \epsilon) \approx \epsilon$$

Thus:

$$H_0 \Delta t \approx 6.96 \times 10^{-10}$$

Using:

$$H_0 \approx 2.2 \times 10^{-18} \text{ s}^{-1}$$

we obtain:

$$\Delta t \approx \frac{6.96 \times 10^{-10}}{2.2 \times 10^{-18}} \approx 3.16 \times 10^8 \text{ s}$$

which corresponds to approximately:

$$\Delta t \approx 10 \text{ years}$$

Equivalent Distance and Redshift

Associating this interval with the light travel distance:

$$L = c\Delta t$$

we obtain:

$$L \approx 9.5 \times 10^{16} \text{ m} \approx 10 \text{ light-years}$$

The corresponding redshift is:

$$z = e^{H_0\Delta t} - 1 \approx 6.96 \times 10^{-10}$$

Final Remark

This comparison shows that the magnitude of gravitational time dilation at the Earth's surface corresponds, within the DUT framework, to an extremely small cosmological effect, equivalent to a redshift of order 10^{-9} or a light-travel distance of approximately 10 light-years.

This highlights that local gravitational time effects are extremely small compared to typical cosmological scales, while still providing a useful reference for comparing the two frameworks.

The **Tully–Fisher relation** is a simple but powerful idea in astrophysics that connects how **fast a galaxy rotates** to how **bright (or massive) it is**.

Intuition (plain language)

Imagine a spinning galaxy like a giant cosmic carousel:

- If it's **very massive**, gravity is stronger → stars must move **faster** to stay in orbit.
- If it's **less massive**, gravity is weaker → stars move **slower**.

Now, brightness (luminosity) is roughly a proxy for how much stuff (stars) the galaxy has. So:

👉 **Brighter galaxies tend to rotate faster.**

The core formula

$$L \propto V^\alpha$$

Where:

- L = luminosity (how bright the galaxy is)
 - V = rotational velocity (how fast it spins)
 - $\alpha \approx 3$ to 4 (depends on the wavelength and data)
-

What this means in practice

If you measure how fast a galaxy rotates (using Doppler shift of its light), you can:

1. Estimate its **true luminosity** using the formula
2. Compare that with how bright it *appears* from Earth
3. Compute its **distance**

👉 That's why the Tully–Fisher relation is widely used as a **distance measurement tool** in cosmology.

[12] The “Decreasing Universe” Articles (full archive)

<https://jocaxx.github.io/DUniverse/Articles.pdf>